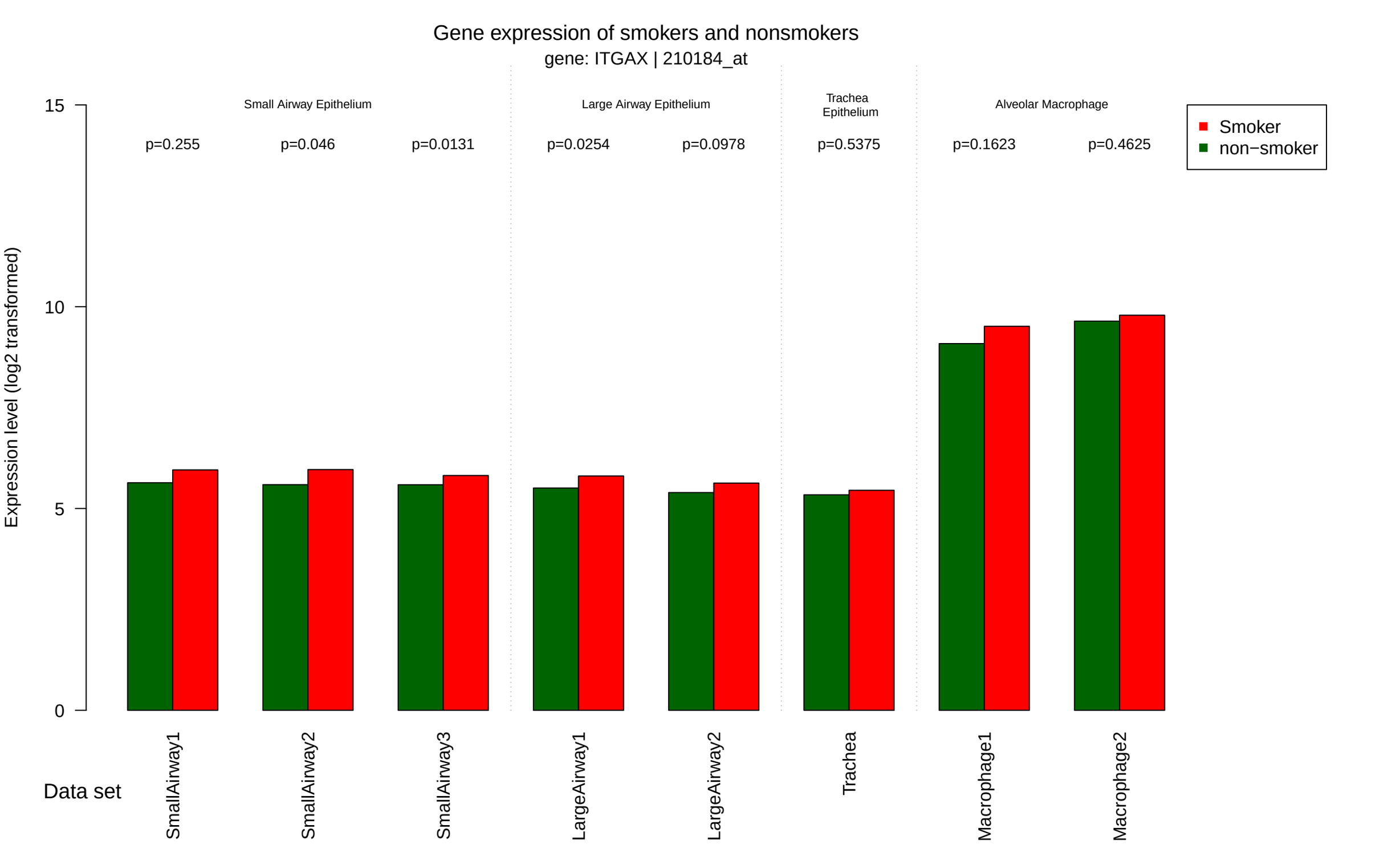
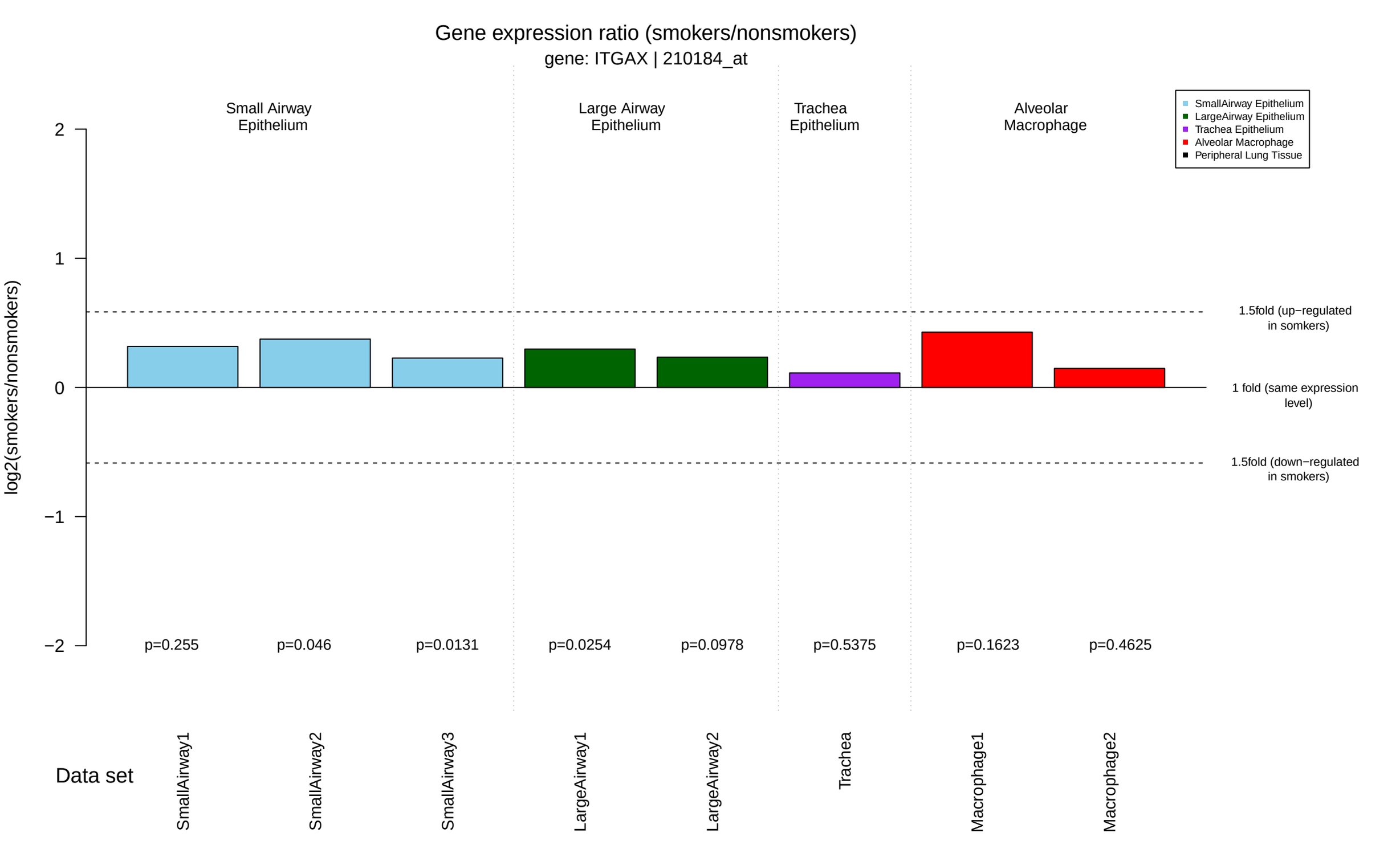
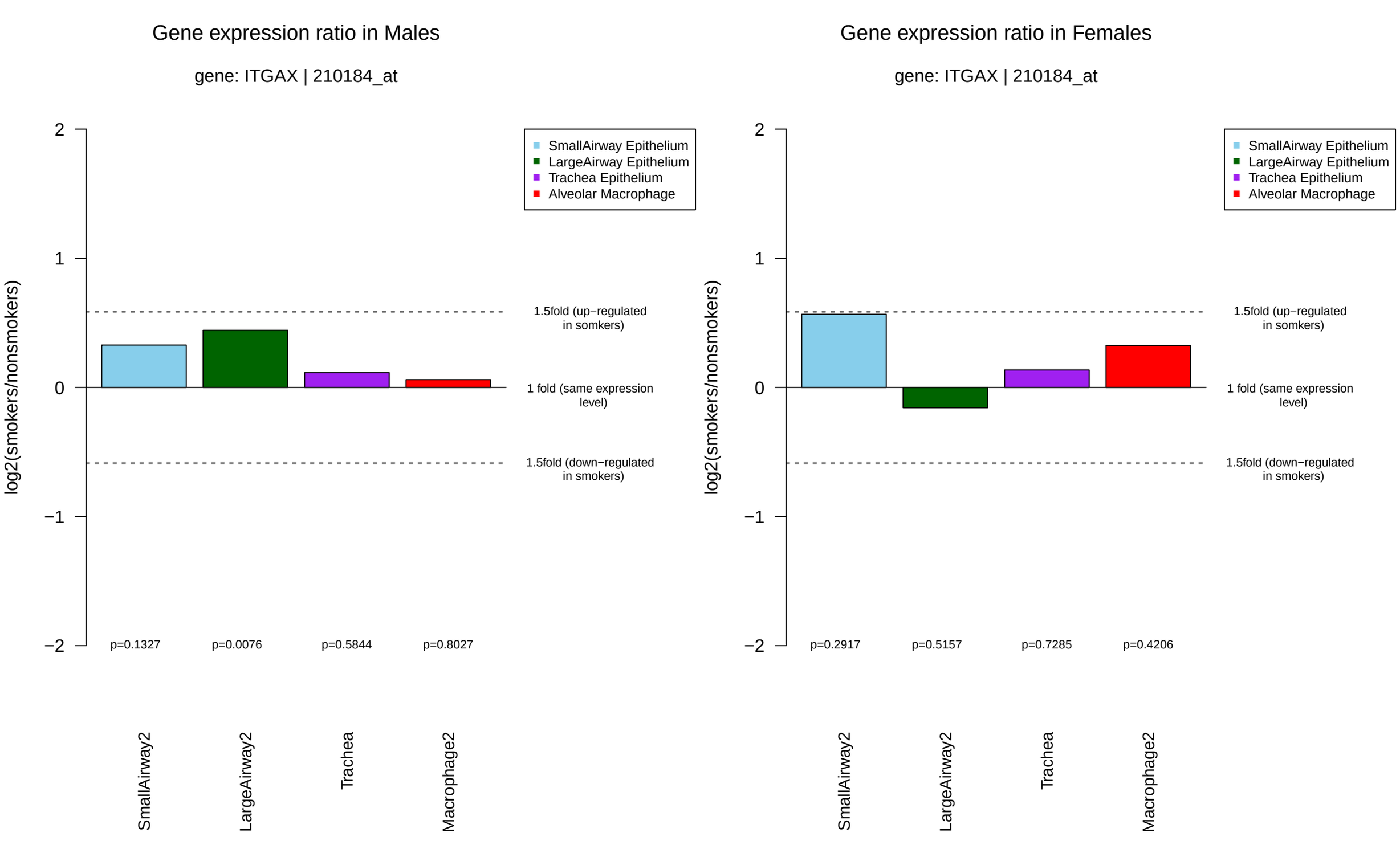
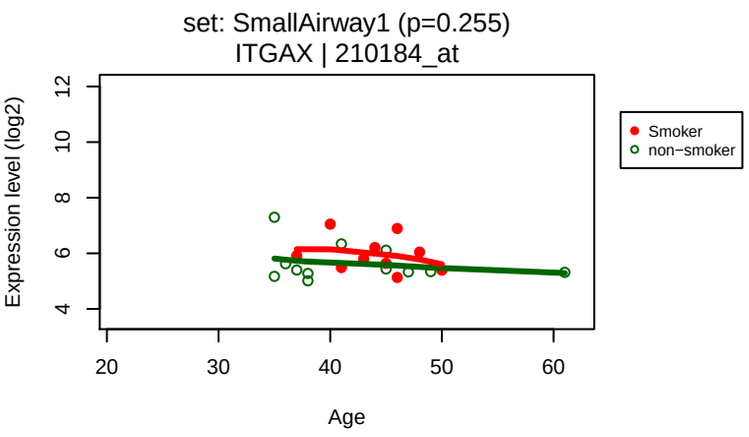
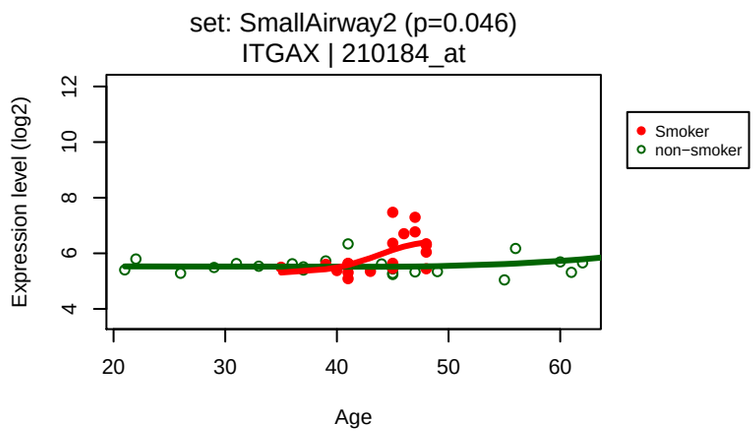
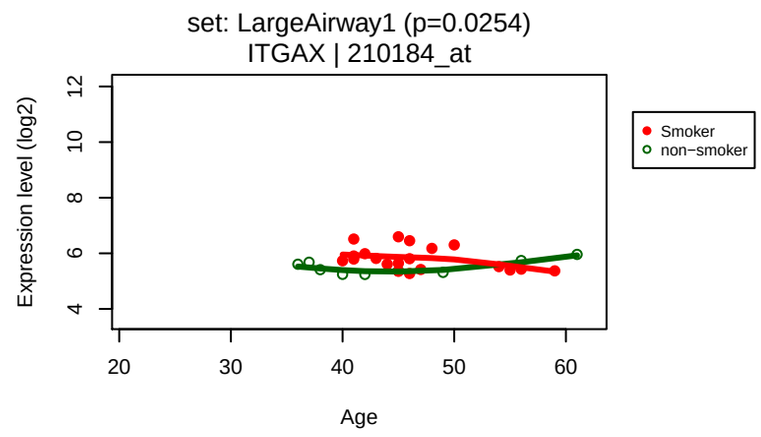

Gene expression of smokers and nonsmokers
gene: ITGAX | 210184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.255
p=0.046
p=0.0131
p=0.0254
p=0.0978
p=0.5375
p=0.1623
p=0.4625
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

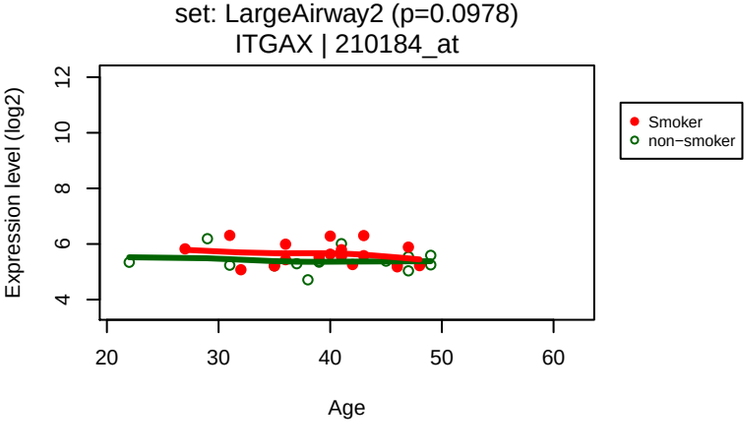

Gene expression ratio (smokers/nonsmokers)
gene: ITGAX | 210184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.255
p=0.046
p=0.0131
p=0.0254
p=0.0978
p=0.5375
p=0.1623
p=0.4625
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

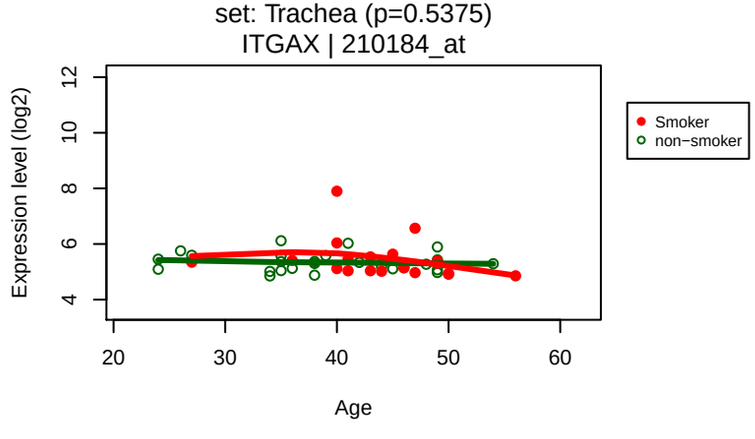

Gene expression ratio in Males
gene: ITGAX | 210184_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1327
p=0.0076
p=0.5844
p=0.8027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGAX | 210184_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2917
p=0.5157
p=0.7285
p=0.4206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

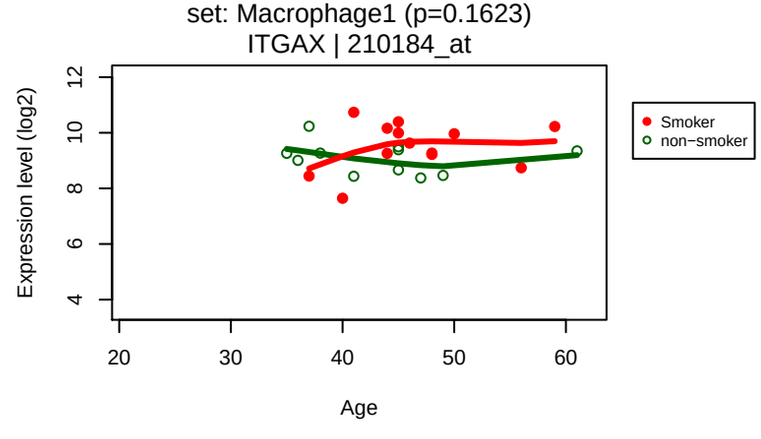

set: SmallAirway1 (p=0.255)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker

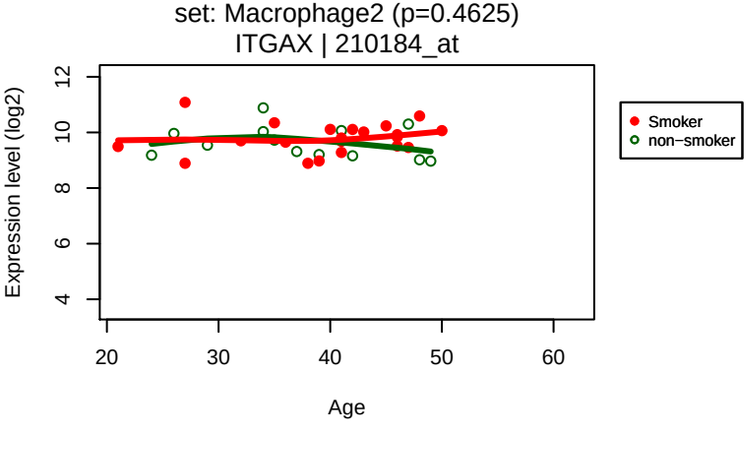

set: SmallAirway2 (p=0.046)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker

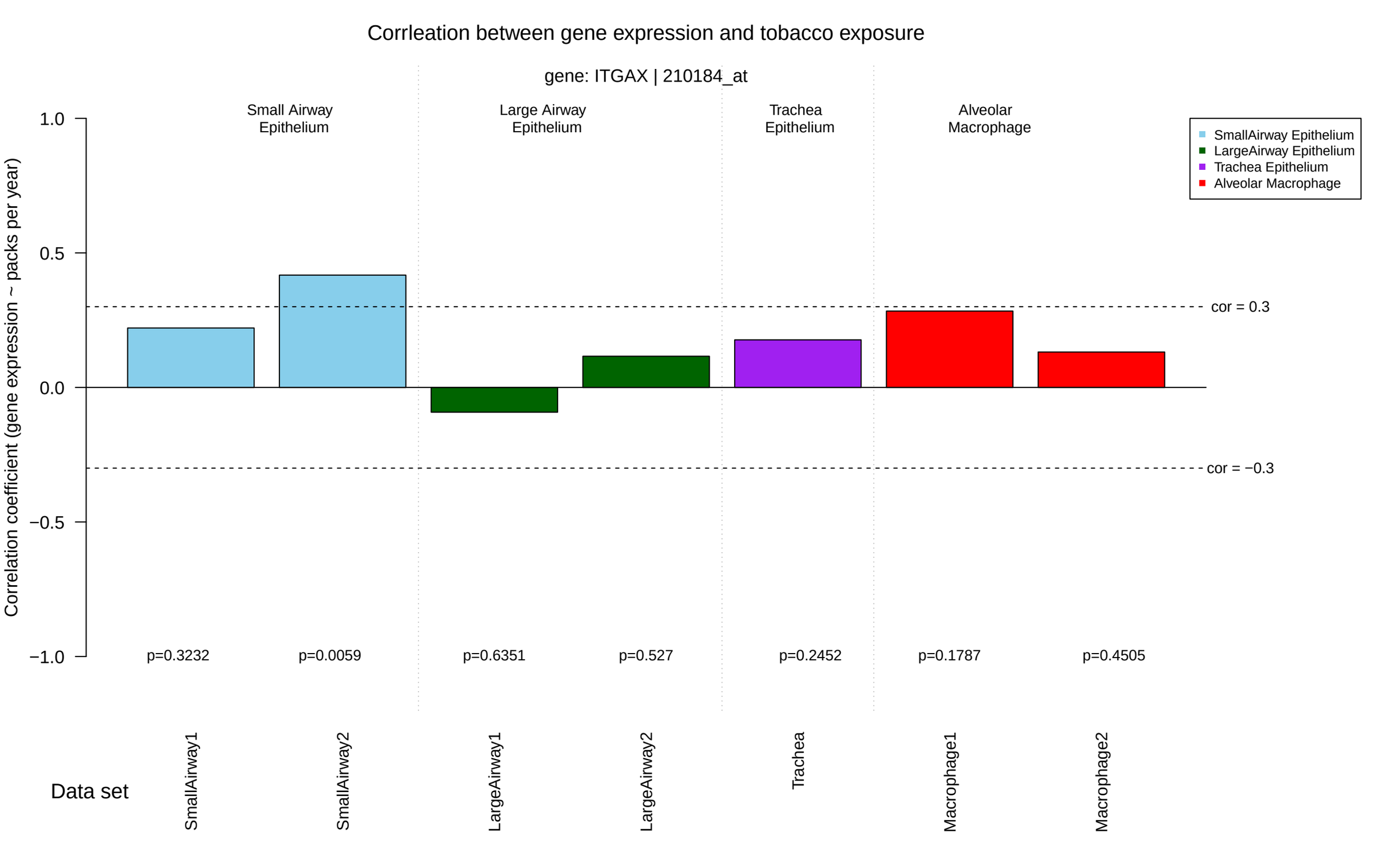

set: LargeAirway1 (p=0.0254)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0978)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5375)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1623)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4625)
ITGAX | 210184_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGAX | 210184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3232
p=0.0059
p=0.6351
p=0.527
p=0.2452
p=0.1787
p=0.4505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2